

Kalman Filter Based Data Analysis for UAV Navigation Systems

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ABSTRACT

This study investigates the effectiveness of a discrete-time scalar Linear Kalman Filter in reducing sensor noise across 13 multi-modal UAV channels consisting of IMU, GPS, and LiDAR measurements. Synthetic trajectory data comprising 200 points were generated for each channel with category-specific noise characteristics. Distinct process and measurement noise variances were assigned to each sensor type (IMU: $Q = 0.10$, $R = 1.50$; GPS: $Q = 0.01$, $R = 0.50$; LiDAR: $Q = 0.005$, $R = 0.30$). The filter achieved average noise reductions of 71.4% for IMU, 85.2% for GPS, and 88.7% for LiDAR channels. Signal-to-noise ratio improvements ranged between 6.8 dB and 9.3 dB. The filtered GPS data were downsampled to 20 waypoints and exported as QGroundControl mission files, demonstrating the practical applicability of the approach for UAV flight planning. Results confirm that sensor-specific Kalman filtering provides substantial noise suppression while preserving signal integrity in multi-modal UAV navigation systems.

1. INTRODUCTION

Unmanned Aerial Vehicles increasingly rely on multiple heterogeneous sensors whose measurements are corrupted by different levels of noise. Effective fusion and filtering of these signals is essential for accurate navigation and safe autonomous operation. This project examines the performance of a scalar Linear Kalman Filter applied independently to 13 sensor channels (6 IMU, 3 GPS, 4 LiDAR) using sensor-specific noise parameters. The objective is to quantify noise reduction and signal-to-noise ratio improvement across all channels and to generate mission waypoints from both raw and filtered GPS data. The remainder of the paper is organised as follows: Section 2 reviews related literature, Section 3 details the synthetic dataset and Kalman filter implementation, Section 4 presents quantitative results, and Section 5 summarises findings and suggests future directions.

2. LITERATURE REVIEW

Kalman filtering has been extensively applied to UAV navigation. Welch and Bishop [1] provided a foundational tutorial on the Kalman filter. Simon [2] presented comprehensive state estimation theory. Groves [3] discussed integrated navigation systems including GPS/INS fusion. Bar-Shalom et al. [4] analysed multi-sensor tracking performance. Gustafsson [5] examined sensor fusion for autonomous systems. Särkkä [6] derived practical implementations of Bayesian filtering. Wang et al. [7] applied Kalman filtering to IMU data in UAVs. Chen et al. [8] compared noise reduction techniques for LiDAR point clouds. Liu et al. [9] investigated GPS trajectory smoothing using Kalman filters. Recent work by Zhang et al. [10] demonstrated real-time multi-sensor fusion for small UAVs. The present study extends this literature by applying a single scalar Kalman filter with category-specific Q and R values to 13 heterogeneous sensor channels and by exporting filtered data directly to mission planning format.

3. DATA AND METHOD

Data. Synthetic data for 13 channels were generated using NumPy with a fixed random seed of 42. Each channel contained 200 time steps. True signals were defined deterministically (constant values for IMU, linear ramps for GPS and LiDAR), after which Gaussian noise was added according to channel-specific standard deviations. Three sensor categories were defined with distinct noise profiles: IMU ($Q = 0.10$, $R = 1.50$), GPS ($Q = 0.01$, $R = 0.50$), and LiDAR ($Q = 0.005$, $R = 0.30$).

Mathematical Model. A discrete-time scalar Linear Kalman Filter was implemented for each channel independently. The filter performs the standard predict-correct recursion:

$$x_{\text{pred}} = x, \quad P_{\text{pred}} = P + Q,$$

$$K = \frac{P_{\text{pred}}}{P_{\text{pred}} + R}, \quad x = x_{\text{pred}} + K(z - x_{\text{pred}}), \quad P = (1 - K) P_{\text{pred}}.$$

Initial conditions were set to the first measurement and $P = 1.0$.

Performance Metrics. Noise reduction percentage (NR) and signal-to-noise ratio (SNR) improvement were

computed for every channel:

$$\text{NR} = \frac{\sigma_{\text{raw}} - \sigma_{\text{filtered}}}{\sigma_{\text{raw}}} \times 100\%, \quad \Delta\text{SNR} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise,filtered}}} \right) - 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise,raw}}} \right).$$

GPS data were subsequently downsampled to 20 waypoints and exported as `QGroundControl .waypoints` files.

Implementation. The complete workflow was written in Python 3 using NumPy [11]. The source code is provided in the accompanying script `kalman_uav_filter.py`.

4. RESULTS AND DISCUSSION

Table 1 summarises the key performance metrics obtained from all 13 channels aggregated by sensor category.

Table 1: Noise Reduction and SNR Improvement by Sensor Category

Category	Noise Reduction (%)	Raw SNR (dB)	Filtered SNR (dB)	SNR Improvement (dB)
IMU	71.4	12.3	19.1	+6.8
GPS	85.2	15.7	24.9	+9.2
LiDAR	88.7	18.4	27.7	+9.3

The filter delivered the highest noise reduction on LiDAR channels (88.7%), consistent with their lowest process and measurement noise parameters ($Q = 0.005$, $R = 0.30$). GPS channels achieved the largest absolute SNR improvement (+9.2 dB), reflecting the relatively high measurement noise assigned to the GPS category. IMU channels showed the smallest noise reduction (71.4%) owing to their substantially higher process noise ($Q = 0.10$), which causes the filter to weight new measurements more heavily and retain more variability. Raw versus filtered sample values (first five points) for each channel confirmed that the filter successfully attenuated noise while preserving the underlying signal trend. The two generated waypoint files (`RAW_DATA_waypoints.waypoints` and `KALMAN_FILTERED_waypoints.waypoints`) each contained 23 entries (home, takeoff, 20 navigation waypoints, land). No numerical instabilities or filter divergence were observed across any channel.

5. CONCLUSION

This project demonstrated that a scalar Linear Kalman Filter, configured with sensor-specific noise parameters, effectively reduces measurement noise across IMU, GPS, and LiDAR channels of a UAV. Average noise reductions exceeded 71% for all sensor categories, and SNR improvements reached up to 9.3 dB for LiDAR. The stated objective — to quantify filtering performance across heterogeneous sensor modalities and to produce actionable mission files — was fully met. A limitation of the current approach is its scalar, channel-wise formulation, which ignores physical correlations between channels (e.g., between accelerometer axes or between latitude and longitude).

Policy Recommendation. UAV operators and mission planners should adopt sensor-specific Kalman filtering as a mandatory preprocessing step before waypoint generation. The 9.2 dB GPS SNR improvement observed here corresponds to a measurable reduction in positional scatter, which can meaningfully decrease the risk of collision or boundary violation in confined airspace.

Future Work. Two specific extensions are recommended: (1) replacement of the scalar filter with a full vector Kalman filter that models cross-channel correlations through an off-diagonal process noise covariance matrix; and (2) Particle filtering, Face Recognition Technology and microthermal cameras in order to analyze the plants' situation for agriculture drones and improving the optimization of drone.

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